

22 Answer: C

When R_{rheo} increases

In the circuit with resistor R,

p.d. V across R_{rheo} increases,
brightness of L1 increases (V^2/R_{L1}).

In the circuit with resistor R_{LDR} ,

p.d. V across R_{rheo} increases,
brightness of L2 increases and more light shines onto LDR
which decreases R_{LDR} , causing V_{rheo} to increase further,
which causes L2 to shine brighter.

Thus L2 is brighter than L1.

When R_{rheo} decreases

In the circuit with resistor R,

p.d. V across R_{rheo} decreases,
brightness of L1 decreases (V^2/R_{L1}).

In the circuit with resistor R_{LDR} ,

p.d. V across R_{rheo} decreases,
brightness of L2 decreases and less light shines onto LDR
which increases R_{LDR} , causing V_{rheo} to decrease further,
which causes L2 to shine less brightly.

Thus L1 is brighter than L2.